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Method and system for estimating the presence of people on a territory exploiting mobile communication network data

Abstract

A method including subdividing a geographic area of interest in geographic zones; subdividing the geographic area of interest in geographic area portions; causing mobile communication terminals situated in the geographic area of interest to calculate respective geographic position estimates; distributing an overall number of geographic position estimates; receiving indications of events of interaction between the mobile communication terminals and network cells; identifying, for each one of the stored indications of events of interaction, the network cell in which the event of interaction occurred; building an array of presences in which each geographic zone has assigned thereto a respective indication of a respective number of events of interaction occurred in the identified network cells which have, in the geographic zone, a cell weight greater than a predetermined cell weight value, and providing the array of presences in output for estimating presence of individuals on a territory.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2006/0293046	12/2005	Smith	N/A	N/A
2012/0115475	12/2011	Miyake	455/435.1	H04W 4/021
2012/0115476	12/2011	Shin et al.	N/A	N/A
2013/0179058	12/2012	Smith	N/A	N/A

OTHER PUBLICATIONS

International Search Report & Written Opinion issued Aug. 18, 2021 in PCT/EP2021/063168 filed on May 18, 2021, 16 pages. cited by applicant

Calabrese et al. “Real-Time Urban Monitoring Using Cell Phones: A Case Study in Rome”, IEEE Transactions on Intelligent Transportation Systems, vol. 12, No. 1, Mar. 2011. pp. 141-151. cited by applicant

Francesco Calabrese and Carlo Ratti, “Real Time Rome”, Networks and Communications Studies, vol. 20, nº 3-4, pp. 247-258, 2006. cited by applicant

Caceres et al. “Exploring the Use of Urban Greenspace through Cellular Network Activity”, 2nd Workshop on Pervasive Urban Applications (PURBA), Jun. 2012. cited by applicant

Deville et al. “Dynamic population mapping using mobile phone data”, PNAS Nov. 11, 2014, 111 (45) 15888-15893; first published Oct. 27, 2014, available at https://doi.org/10.1073/pnas.1408439111. cited by applicant
Danya Bachir, “Estimating Urban Mobility with Mobile Network Geolocation Data Mining”, Université Paris Saclay; https://theses.hal.science/tel-02046122v1, Feb. 22, 2019. cited by applicant

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Background/Summary

BACKGROUND OF THE PRESENT DISCLOSURE
Field of the Present Disclosure
(1) The present disclosure generally relates to the field of mobile communications and mobile communication networks. More specifically, the present disclosure relates to a method and a system for exploiting data of a mobile communication network for estimating presences of individuals on a territory.
Overview of the Related Art
(2) Mobile phones (also referred to as cell phones, hand phones, mobile terminals, mobile stations, user equipment, and, for the purposes of the present disclosure, intended to include smartphones and tablets and, more generally, mobile communication devices with 2G-3G-4G-5G connectivity) have become an essential part of the everyday life of many people. Due to this, it has been realized that mobile phones can expediently be used as sensors for monitoring the behaviour of communities of persons in several fields of application.
(3) WO 2013/107669 A1 discloses a method to automatically detect and label one or more points of interest (Pols) of a user of mobile phone services, said method being based exclusively on geo-located phone usage information and with no customer interaction. Based on the geo-located usage events generated in the telecommunications operator network and using statistical methods, the method allows the identification of, from all the locations visited by the user, the locations that are most relevant to him: his Pols. The method automatically assigns labels to the Pols detected, thereby attaching meaning to such locations.
(4) US 2006/0293046 proposes a method for exploiting data from a wireless telephony network to support traffic analysis. Data related to wireless network users are extracted from the wireless network to determine the location of a mobile station. Additional location records for the mobile station can be used to characterize the movement of the mobile station: its speed, its route, its points of origin and destination, and its primary and secondary transportation analysis zones. Aggregating data (http://en.wikipedia.org/wiki/Voronoi_diagram). Such a Voronoi tessellation is achieved starting from the locations of the radio base stations of the mobile communication network and taking into account the directionality of the antennas installed on the radio base stations. To each tile of the tessellation a network cell is univocally associated and, consequently, each tile is assigned the whole value of the data available at the cell level. Thus, a uniform distribution of the data available at the cell level onto the corresponding tile is achieved.
(5) The following works provide a step forward in the calculation of the presence of individuals on an arbitrary partition in zones of a geographic area, instead that on pixels or on areas linked to the

mobile communication networks.
(6) In Pierre Deville, Catherine Linard, Samuel Martin, Marius Gilbert, Forrest R. Stevens, Andrea E. Gaughan, Vincent D. Blondel, and Andrew J. Tatem, "Dynamic population mapping using mobile phone data", PNAS Nov. 11, 2014 111 (45) 15888-15893; first published Oct. 27, 2014, available at https://doi.org/10.1073/pnas. 1408439111, it is shown how spatially and temporarily explicit estimations of population densities can be produced at national scales, and how these estimates compare with outputs produced using alternative human population mapping methods. It is also demonstrated how maps of human population changes can be produced over multiple timescales while preserving the anonymity of MP users. The total number of users of the j-th cell, Tj,is distributed onto any i-th area ai by: 1) considering the Voronoi polygon vj corresponding to the j-th cell (calculated in standard way as described in http://en.wikipedia.org/wiki/Voronoi_diagram); 2) distributing the data Tj uniformly over the polygon vj; 3) detecting the intersection between the polygon and the area ai, and then 4) uniformly distributing the users on such intersection.
(7) Consequently, the wider the intersection the greater the quota of the data Tj distributed on the area ai.
(8) In Danya Bachir, "Estimating Urban Mobility with Mobile Network Geolocation Data Mining", Thèse de doctorat de l'Université Paris-Saclay préparée à IRT SystemX, Bouygues Telecom, Télécom SudParis, Ecole doctorale n°1 Sciences et technologies de l'information et de la communication (STIC); Spécialité de doctorat: Réseaux, information et communications; thesis presented and discussed at Gif-sur-Yvette on 25 Jan. 2019, urban mobility is estimated by mining mobile network data, which are collected in real-time by mobile phone providers at no extra-cost. Processing the raw data is reported to be non-trivial as one must deal with temporal sparsity, coarse spatial precision and complex spatial noise. The thesis work addresses two problematics through a weakly supervised learning scheme (i.e., using few labelled data) combining several mobility data sources. First, population densities and number of visitors over time are estimated, at fine spatio-temporal resolutions. Second, Origin-Destination matrices are derived representing total travel flows over time, per transport modes. All estimates are exhaustively validated against external mobility data, with high correlations and small errors. The distribution of the total number of users Tj of the j-th cell ("total number of different users Tj") is distributed on any i-th area ai taking into account two factors: on one side, as in the preceding work, there is an intersection between the Voronoi polygon of the j-th cell and the area ai, on the other side there is the "census population" in the area ai: the greater the population density, the greater the quota of Tj that is assigned to the area ai (in the calculation, use is made of the population density assuming a uniform distribution on the area).
SUMMARY OF THE PRESENT DISCLOSURE
(9) The Applicant has observed that the known methods for estimating presence of individuals on a territory are not satisfactory.
(10) In all the works discussed in the technical literature outlined in the foregoing, methods for calculating the presence of individuals on a territory are proposed that start from data extracted from the mobile communication network, and that, through simple and easy to implement software methods, distribute the aggregated data at the network cell level on the territory. However, at the same time, such methods are based on a hypothesis of uniform/exponential distribution of population that is unrealistic: in everyday life the users of the mobile communication network are not uniformly/exponentially distributed on the territory with respect to network cells or radio sites. This causes gross errors when one is interested in knowing the presence of individuals at a local level, at a scale (much) smaller than that of the network cell size, as requested by the practical applications mentioned before.
(11) The Applicant has observed that in several activities of planning and managing of services offered to the population, particularly activities of management of emergencies and security

services, as well as activities of marketing and promotion (e.g., poster advertising), and for monitoring events and estimating the number of attendees, it is important to accurately calculate the presence of individuals on the territory at any given time, on any possible geographic area of interest.

(12) Contrary to this, as mentioned a few lines above, works available in the technical literature disclose methods that are unrealistic and source of gross errors.

(13) The Applicant believes that there is the need of overcoming these problems by defining a new method for calculating the presence of people on the territory, that reflects the actual positioning of individuals and that can be used for estimating the presence of people at a scale even much smaller than the typical size of mobile communication network cells.

(14) The Applicant has found that this goal can be achieved starting from a certain number of estimates of position that a (properly selected) set of mobile terminals perform and send to a central system (apparatus or server). These position estimates, properly processed, allow calculating weights to be assigned to cells of the mobile communication network on zones of a subdivision in zones of an area of interest. Such weights are indications of the probabilities that an individual which is a user of a mobile terminal located in an area covered by each network cell is also located in a specific zone among the zones in which the area of interest is subdivided.

(15) Through such weights, data available at and provided by the mobile communication network is distributed through the zones to obtain a count of the presence of individuals on each zone of the area of interest in a certain time interval.

(16) The position estimates performed and provided by the mobile communication terminals can be obtained preliminarily to the count of the presence of individuals and can be re-used more times in the count of the presence of individuals for an area of interest thereby achieving an efficient use of mobile communication network resources and a reduction of computational burden.

(17) By “data available at the mobile communication network” it is meant data indicating events of interaction between mobile phones of users located in the area of interest and the cells of the mobile communication network. Events of interaction useful for the purposes of the present disclosure include for example one or more among (the following list is not exhaustive and not limitative, and may for example change with the evolution of the mobile communications technologies): interactions at the power-on and/or power-off of the mobile phones; interactions at location area update (either periodical, performed with a periodicity imposed by the mobile communication network, or non-periodical, carried out when a mobile communication device informs the cellular network whenever it moves from one “Location Area” to the next; as known, a “Location Area” is a portion of a territory formed by a group of network cells); interactions at the re-entrance of the mobile phone into a covered area, covered by the mobile communication network, from an area not covered by the network; interactions at the switch of the mobile phone from a mobile network technology to another (e.g., from 2G to 3G or viceversa, etc.); interactions at incoming/outgoing calls (namely, at placing an outgoing call, at answering an incoming call); interactions at sending/receiving SMS and/or MMS; interactions at Internet access (from a browser or any other app of the mobile phone).

(18) According to an aspect of the present disclosure, there is provided a method, implemented by a data processing system, for estimating presence of individuals on a territory based on data derived from a mobile communication network comprising a plurality of network cells covering said territory, wherein said individuals are users of mobile communication terminals configured to be adapted to interact with the mobile communication network and located in said territory.

(19) The method comprises: subdividing a geographic area of interest in geographic zones; subdividing the geographic area of interest in geographic area portions; causing mobile communication terminals served by the mobile communication network and situated in the geographic area of interest to calculate respective geographic position estimates and provide the calculated geographic position estimates to the data processing system; distributing an overall

number of geographic position estimates, received from the mobile communication terminals, on the geographic area portions by assigning to each geographic area portion a respective number of geographic position estimates corresponding to geographic positions estimates falling within said geographic area portion; for each of said at least one network cell: determining, among said geographic area portions, covered geographic area portions falling within a coverage area of said network cell; assigning to each one of the determined covered geographic area portions a respective weight which depends on said respective number of geographic position estimates compared to an overall number of geographic position estimates falling within all said determined covered geographic area portions; generating a correspondence map establishing a correspondence between said network cells and said geographic zones, said generating the correspondence map comprising, for each of said network cells, calculating a cell weight of the network cell on each geographic zone by determining the geographic area portions belonging to each of said zones and summing the respective weights of the geographic area portions identified as belonging to said each of said zones; receiving from the mobile communication network and storing in a repository indications of events of interaction between the mobile communication terminals and the network cells of the mobile communication network; identifying, for each one of the stored indications of events of interaction, the network cell in which the event of interaction occurred; using the correspondence map, building an array of presences in which each of said geographic zones has assigned thereto a respective indication of a respective number of events of interaction occurred in the identified network cells which have, in said each of said geographic zones, a cell weight greater than a predetermined cell weight value, and providing said array of presences in output to said data processing system for the use, by a user of the data processing system, for estimating presence of individuals on a territory.
(20) In embodiments, the events of interaction may comprise at least one among: interactions at the power-on and/or power-off of the mobile communication terminals; interactions at location area update procedures; interactions at the re-entrance of the mobile communication terminals into a covered area, covered by the mobile communication network, from an area not covered by the network; interactions at the switch of the mobile communication terminals from a mobile network technology to another; interactions at incoming/outgoing calls; interactions at sending/receiving SMS and/or MMS; interactions at Internet access.
(21) In embodiments, the building an array of presences may comprise building a set of arrays of presences, each array of presences of said set being related to a respective time slot of a plurality of time slots in which an observation time period is subdivided.
(22) In embodiments, the causing mobile communication terminals served by the mobile communication network and situated in the geographic area of interest to calculate respective geographic position estimates and provide the calculated geographic position estimates to the data processing system may comprise selecting, among all the mobile communication terminals served by the mobile communication network and situated in the geographic area of interest, a subset of selected mobile communication terminals according to a selection criterion comprising at least one among: mobile communication terminals of users which have given their consent to activate their geographic localization; mobile communication terminals of users which have given their consent to take part to measurement campaigns to be performed for purpose of calculating presence of individuals on a geographic territory; mobile communication terminals of users which, at subscription, have declared a residence in a certain area inside the geographic area of interest; mobile communication terminals of users in respect of which the operator of the mobile communication network has determined a residence or a working location in a certain area inside the geographic area of interest.
(23) In embodiments, the method may comprise configuring the subset of selected mobile communication terminals, said configuring comprising providing to the selected mobile communication terminals at least one among: an indication of a time interval during which the

selected mobile communication terminals will have to calculate the respective geographic position estimates; an indication of a periodicity with which the selected mobile communication terminals will have to calculate the respective geographic position estimates; an indication of whether the selected mobile communication terminals will have to provide to the data processing system the respective calculated geographic position estimates immediately after the respective geographic position estimates have been calculated or the respective calculated geographic position estimates can be provided to the data processing system at a later time after they have been calculated.
(24) In embodiments, the determining covered geographic area portions falling within a coverage area of that network cell may comprise: obtaining a coverage area of that network cell; determining which of said geographic area portions belong to the coverage area of the network cell, wherein said determining comprises at least one among: assessing which of said geographic area portions have geographic coordinates of a center of gravity thereof falling inside the coverage area of the network cell; assessing which of said geographic area portions fall, for a given percentage of their area, inside the coverage area of the network cell.
(25) In embodiments, the determining the geographic area portions belonging to each of said zones may comprise one among: assessing which of said geographic area portions have geographic coordinates of a center of gravity thereof falling inside the zone; assessing which of said geographic area portions fall, for a given percentage of their area, inside the zone.
(26) In embodiments, the assigning to each geographic area portion a respective number of geographic position estimates corresponding to geographic positions estimates falling within said geographic area portion may comprise, in case a geographic position estimate falls on a boundary between at least two geographic area portions: either randomly selecting one of the at least two geographic area portions to be assigned to that geographic position estimate, or assigning to each of said at least two geographic area portions a fraction of said geographic position estimate equal to 1 divided by the number of said at least two geographic area portions.
(27) In embodiments, the calculated geographic position estimates may be provided in the form of t-uples, wherein each t-uple may comprise: a field containing an indication of the geographic position estimated by the mobile communication terminal, expressed in any system of geographic coordinates, a field containing an indication of the time instant at which the geographic position estimate has been calculated, optionally a field containing an indication of a univocal identifier of the mobile communication terminal that calculated and sent the geographic position estimate.
(28) According to another aspect of the present disclosure, a data processing system is provided comprising modules configured to perform, when operated, the method of the previous aspect.
(29) Thanks to the solution according to the present disclosure, the presence of people on the territory can be calculated in a way that reflects the actual positioning of individuals, even at a scale much smaller than the typical size of mobile communication network cells.

Description

BRIEF DESCRIPTION OF THE ANNEXED DRAWINGS
(1) These and other features and advantages of the solution according to the present disclosure will be better understood by reading the following detailed description of embodiments thereof, provided merely by way of non-limitative examples, to be read in conjunction with the attached drawings, wherein:
(2) FIG. **1** schematically depicts a system according to an embodiment of the present disclosure for calculating presence of individuals on a geographic territory;
(3) FIG. **2** schematically depicts a component (referred to as Localization Manager) of the system of FIG. **1**;
(4) FIG. **3** schematically depicts a surveyed geographic area, hereafter also referred to as “Region

of Interest" ("RoI" in short), and a subdivision thereof in zones;
(5) FIG. **4** shows an Array of Presences generated by the system of FIG. **1**, containing indications of the calculated numbers of individuals present in each of the zones of the RoI;
(6) FIG. **5** shows a set of M Arrays of Presences calculated by the system of FIG. **1** in respect of M time slots in which an observation time period is subdivided;
(7) FIG. **6** schematically shows a tessellation of the RoI;
(8) FIG. **7** shows estimates (measurements) sent by User Equipment ("UE" in short) operable in a mobile communication network to the Localization Manager, containing estimates, calculated by the UE, of their geographic positions, together with associated timestamps;
(9) FIG. **8** shows a high-level flowchart of the operations, carried out by the system of FIG. **1**, for constructing cells-zones correspondence maps;
(10) FIG. **9** shows a more detailed flowchart of a step of UE configuration of the flowchart of FIG. **8**;
(11) FIG. **10** shows a more detailed flowchart of a step of general distribution map construction of the flowchart of FIG. **8**;
(12) FIG. **11** shows a more detailed flowchart of a step of cell-specific distribution maps construction of the flowchart of FIG. **8**;
(13) FIG. **12** shows a more detailed flowchart of a step of construction of cells-zones correspondence maps of the flowchart of FIG. **8**, and
(14) FIG. **13** shows a detailed flowchart of the operations carried out by the system of FIG. **1** for calculating the presence of individuals on the RoI and for building the Arrays of Presences.
DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS OF THE PRESENT DISCLOSURE
(15) With reference to the drawings, FIG. **1** is a schematic representation of a system **100**, according to an exemplary embodiment of an aspect of the present disclosure, for calculating presence of individuals on a geographic territory.
(16) The system **100** is coupled with a mobile communication network **105**, such as a (2G, 3G, 4G, 5G or higher/other generation) mobile telephony network deployed on and covering a geographic area (in the following also referred to, shortly, as area or surveyed area or surveyed geographic area) pictorially delimited by and within the dash-and-dot line **107**. The area **107** can for example be a city, a municipality, a district, etc.
(17) The system **100** is configured to receive from the mobile communication network **105**, for (each or a subset of) UE of individuals located in the surveyed area **107**, respective positioning data (i.e., geolocation data). The UE, not shown in the drawings, can be for example mobile phones, smartphones, tablets, with 2G-3G-4G-5G or higher/other connectivity, etc., The mobile communication network **105** comprises a plurality of (two or more) communication stations **105***a* (e.g., radio base stations of the mobile communication network **105**) geographically distributed in the surveyed area **107**. Each communication station **105***a* is adapted to manage communications of UE in one or more served areas or network cells **105***b* (in the example at issue, three cells are assumed to be served by each communication station **105***a*).
(18) Each radio base station **105***a* of the mobile communication network **105** is configured to be adapted to interact with any UE located within one of the cells **105***b* served by such communication stations **105***a* (e.g., interactions include interactions at UE power on/off, at location area update, at incoming/outgoing calls, at sending/receiving SMS and/or MMS, at Internet access, etc.). In this document, such interactions between the UE and the mobile communication network **105** are generally referred to as "events" e.sub.v (with v=1, . . . , V; V>0).
(19) The surveyed area **107** may be regarded as subdivided in a plurality of sectors, each one corresponding to a respective cell **105***b* of the (part of the) mobile communication network **105** superimposed on the surveyed area **107**.
(20) The system **100** comprises a Computation Engine **110** configured to be adapted to process data

retrieved from the mobile communication network **105**. The system **100** comprises a Repository **115** (such as a database, a file system, etc.) configured to be adapted to store data regarding the interactions between the UE and the mobile communication network **105**. The Repository **115** can be configured to be adapted to store computation results generated by the Computation Engine **110**. The Repository **115** can be configured to be adapted to store any processing data generated by, and/or provided to, the system **100** (generally in a binary format).

(21) The data stored in the Repository **115** include data about the coverage (coverage area data) of the cells **105***b* of the mobile communication network **105**. For the purposes of the present disclosure, the way in which the coverage area data are expressed is not relevant: the coverage area of a cell can indifferently be expressed by way of the geographic coordinates of the center and the radius of a circle defining the cell coverage area or by way of areas delimited by polygons, regular or irregular, with any number of sides, or by way of the union of square pixels, joint or separated. Additionally, the present disclosure does not depend on the way adopted for the computation of the cells' coverage area, so that it is possible to exploit, indifferently, both simplified electromagnetic field propagation models (such as those described in Theodore S. Rappaport, "Wireless Communications", Prentice Hall, 1996, Chapter 3 and Chapter 4, pages 69-196) and mobile network planning tools normally exploited by mobile communication network operators. Moreover, the coverage of the network cells can indifferently be calculated specifically, ad-hoc for the system **100** before the start of the operation of the system **100**, or be coverage area data already available to the operator of the mobile communication network **105** for the network planning.

(22) The system **100** is provided with an Administrator Interface (A.I.) **120** (e.g., implemented in a computer) configured and operable for modifying parameters and/or algorithms used by the Computation Engine **110** and/or accessing data stored in the Repository **115**.

(23) The system **100** may comprise one or more User Interface(s) (U.I.) **125** (e.g., a user terminal, or a software running on a remote terminal connected to the system **100**) configured to be adapted to receive inputs from, and to provide output to a user of the system **100**. In the present disclosure the expression "user of the system" may refer to one or more human beings and/or to external computing systems (such as a computer network, not shown) of a third party being subscriber of the services provided by the system **100** and enabled to access the system **100**—e.g., under subscription of a contract with a service provider owner of the system **100**, and possibly with reduced right of access to the system **100**, limited compared to the right of access held by an administrator of the system **100** operating through the Administrator Interface **120**. The system **100** comprises a Localization Manager **130**, shown in greater detail as **200** in FIG. **2**, from a logical viewpoint. The Localization Manager **200** comprises a Communication Manager **205** and a Position Manager **210**. The Communication Manager **205** is configured to be adapted to manage the communications with the UE (possibly, with a subset of all the UE) that happen to be located in the surveyed area **107**, particularly for transferring to the UE configuration parameters of the system (as described in detail later on). The Position Manager **210** is configured to be adapted to receive position estimates, with related timestamps, calculated and sent by the UE, and to transfer the received position estimates and timestamps, when needed, to the Computation Engine **110**, for processing by an algorithm for the calculation of presence of individuals on the territory.

(24) The Localization Manager **200** can also include a Position Repository **215** (such as a database, a file system, etc.) in which the position estimates (and related timestamps) provided by the UE are stored. The Position Repository **215** can also be physically integrated in the Repository **115**, if suitable (e.g., in case the Localization Manager **200** is physically integrated with the other components of the system **100** or when the transfer of data between the Localization Manager **200** and the Repository **115** can be accomplished quickly without causing delays on the processing operations).

(25) The Localization Manager **200** is configured to be adapted to communicate with the UE through the mobile communication network **105**. However, for the purposes of the present

disclosure the specific way in which the communication between the Localization Manager **200** and the UE is accomplished, and the protocols adopted therefore, are not limitative. The dialogue between the Localization Manager **200** and the UE can for example take place through the User-plane, e.g., using communication protocols of the Internet world (like the TCP/IP), in a way totally similar to what normally happens between a terminal and a remote server, or through the Control-plane, encapsulating the messages exchanged between the Localization Manager **200** and the UE in signaling messages that apparatuses of the mobile communication network **105** exchange for performing their specific operations.

(26) The system **100** can be implemented in any known manner; for example, the system **100** can comprise a single computer, or a network of distributed computers, either of physical type (e.g., with one or more main machines implementing the Computation Engine **110** and the Repository **115** connected to other machines implementing Administrator and User Interfaces **120** and **125**) or of virtual type (e.g., by implementing one or more virtual machines in a network of computers).

(27) Additionally, some modules and/or functionalities of the system **100**, such as the Localization Manager **200**, can be part of, implemented in, the mobile communication network **105** or possibly be implemented in some apparatuses of the mobile communication network **105** or in the O&M system thereof. The different functionalities of the Localization Manager **200** can be implemented in different apparatuses, depending on the specific necessities of the mobile communication network operator and/or administrator of the system **100** (costs, availability of apparatus, available communication bandwidth for the links, etc.). For the purposes of the present disclosure, these aspects of implementation detail are not strictly relevant.

(28) The system **100** is configured to be adapted to retrieve (and/or receive) from the mobile communication network **105** an event record er.sub.v (v=1, . . . , V; V positive integer) for each event e.sub.v (v=1, . . . , V; I′ positive integer) occurred between a generic UE and the mobile communication network **105** (through one of its communication stations **105***a*) within the surveyed area **107**. Each event record er.sub.v retrieved by the system **100** from the mobile communication network **105** can comprise—in a non-limitative manner—an identifier of the UE that is involved in the corresponding event e.sub.v (e.g., the UE identifier may be selected as one or more among the International Mobile Equipment Identity-IMEI, the International Mobile Subscriber Identity-IMSI and the Mobile Subscriber ISDN Number-MSISDN code), time data (also denoted as timestamps) indicating the time at which the corresponding event e.sub.v has occurred, and an identifier of the cell **105***b* involved in such event e.sub.v.

(29) In one embodiment, the UE identifier of the UE involved in the event record er.sub.v′ may be provided not in clear, e.g. as an encrypted information in order to ensure the privacy of the UE owner. In other words, if the need arises, the information (i.e., the identity of the owner of the UE corresponding to the UE identifier) may be encrypted/decrypted by implementing a suitable encryption/decryption algorithm, or a cryptographic hash function such as for example the algorithm SHA256 described in “Secure Hash Standard (SHS)”, National Institute of Standards and Technology FIPS—180-4, Mar. 6, 2012. In this way, the identity of the individual is not in clear, being replaced by a non-intelligible pseudonym.

(30) The system **100** may retrieve (and/or receive) the event records er.sub.v related to a generic UE from the mobile communication network **105** by acquiring records of data generated and used in the mobile communication network **105**. For example, in case the mobile communication network **105** is a GSM network, Charging Data Records (CDRs), also known as call data records, and/or Visitor Location Records (VLRs) may be retrieved from the mobile communication network **105** and reused as event records er.sub.v by the system **100** for its purposes. The CDR is a data record (usually used for billing purposes by a mobile telephony service provider operating through the mobile communication network **105**) that contains attributes specific to a single instance of a phone call or other communication transaction performed between a UE and the mobile communication network **105**. The VLRs are databases listing UE that have roamed into the

jurisdiction of a Mobile Switching Center (MSC, not shown) of the mobile communication network **105**, which is a management element of the mobile communication network **105** managing events over a plurality of communication stations **105*a***. Each communication station **105*a*** in the mobile communication network **105** is usually associated with a respective VLR.
(31) Conversely, if the mobile communication network **105** is an LTE network, records of data associated with the event records er.sub.v of a generic UE are generated by a Mobility Management Entity (MME), comprised in the mobile communication network **105**, which is responsible for a UE tracking and paging procedure in LTE networks (where no VLR is implemented).
(32) It should be noted that the method described in the present disclosure may be implemented by using any source of data (e.g., provided by one or more WiFi networks) from which it is possible to obtain event records er.sub.v comprising a univocal identifier of (mobile terminals of) individuals (such as the UE identifier mentioned above), a location position of such individuals, and a time indication of an instant during which such event has occurred.
(33) In operation, event records er.sub.v may be continuously retrieved by the system **100** from the mobile communication network **105**. Alternatively, event records er.sub.v may be collected by the system **100** periodically, e.g. for a predetermined time period (e.g., every certain number of hours, or on a daily or weekly basis). For example, event records er.sub.v may be transferred from the mobile communication network **105** to the system **100** as they are generated, in a sort of "push" modality, or event records er.sub.v may be collected daily in the mobile communication network **105** and then packed and transferred to the system **100** periodically or upon request by the system **100**.
(34) The event records er.sub.v retrieved from the mobile communication network **105** are stored in the Repository **115**, where they are made available to the Computation Engine **110** for processing. Event records er.sub.v generated by a same UE may be grouped together in the Repository **115**, i.e. event records er.sub.v may be grouped together if they comprise a common UE identifier and are in this case denoted as event records group erg.sub.n (e.g., n=0, . . . , N, N≥0) hereinafter.
(35) The Computation Engine **110** can be configured to be adapted to execute an algorithm (described in the following) for counting presence of individuals on a geographic area, implemented by a software program product stored in a Memory Element **110*a*** of the system **100**, comprised in the Computation Engine **110** in the example of FIG. **1**, even though the software program product could be stored in the Repository **115** as well (or in any other memory element, not shown, provided in the system **100**).
(36) The event records er.sub.v can be processed according to (as discussed in detail in the following) instructions provided by the system administrator (through the Administrator Interface **120**), for example stored in the Repository **115**, and, possibly, according to instructions provided by a user of the system **100** (through the User Interface **125**).
(37) The Computation Engine **110** may provide the results of the processing performed on the event records er.sub.v to the user of the system **100** through the User Interface **125**, and, optionally, the Computation Engine **110** may be configured to store such processing results in the Repository **115**.
(38) The system **100** might be adapted to retrieve (or receive) data about individuals not exclusively from a mobile communication network like the mobile communication network **105**. Alternatively or in addition, the system **100** may be configured to retrieve (or receive) data about individuals from one or more wireless computer networks, such as WLANs, operating in the surveyed area **107**, provided that the UE of the individuals are capable to connect to such wireless computer networks and that the UE (at least a subset of all the UE located in the surveyed area **107**) are capable of estimating their position by means of any geolocation technique (such as the GPS).
(39) FIG. **3** schematically depicts the surveyed geographic area **107**, in the following referred to as Region of Interest (RoI) and identified as **300**, as the area within the ellipsis delimited by a boundary, or external cordon **305** (which is assumed to delimit the RoI **300**). It is pointed out that

the elliptical shape of the RoI **300** is merely exemplary and not limitative: any other shape is possible.
(40) The RoI **300** is a geographic region within which the presence of individuals are to be calculated by the system **100**. For example, the RoI **300** may be either a district, a town, a city, or any other kind of geographic area. Moreover, the RoI **300** may comprise a number of sub-regions including non-adjacent geographical locations, such as for example a plurality of different cities, different counties and/or different nations (and so on). In addition, the RoI **300** may comprise a set of one or more predetermined locations (such as for example airports, bus/train stations, etc.).
(41) The RoI **300** is subdivided into a plurality of zones, or simply zones z.sub.q (q=1, . . . , Q, where Q is an integer number, and Q>0) in which it is desired to count the presence of individuals. In the example shown in FIG. **3**, the RoI **300** is subdivided into nine zones z.sub.1, . . . , z.sub.9 (i.e., Q=9).
(42) Each zone z.sub.q may be determined by using a zoning technique. According to this zoning technique, each zone z.sub.q may be delimited by administrative (city limits, National boundaries, etc.) and/or physical boundaries (such as rivers, railroads etc.) within the RoI **300** that may hinder the traffic flow and may comprise adjacent lots of a same kind (such as open space, residential, agricultural, commercial or industrial lots) which are expected to experience similar traffic flows. The zones z.sub.q may differ from one another in size. Anyway, it is pointed out that the solution according to embodiments of the solution of the present disclosure is independent from the criteria used to partition the RoI **300** into zones z.sub.q.
(43) FIG. **4** shows an Array of Presences **400** that is calculated by the system **100** on the RoI **300**. The Array of Presences **400** is referred to a respective time interval or time slot ts of an observation time period TP, as described in greater detail in the following. The generic Array of Presences **400** comprises Q rows i and one column. Each row i is associated with a corresponding zone z.sub.q of the RoI **300**; thus, considering the RoI **300** in the example of FIG. **3**, one of the corresponding Array of Presences **400** comprises nine rows i=1, . . . , 9. Each generic element vp(i) of the Array of Presences **400** represents the number of presences in the zone z.sub.t in the corresponding time slot ts.
(44) As outlined above, in order to obtain a more detailed analysis of the presence of individuals, a predetermined observation time period TP of the presence analysis in the region of interest can also established and it can be subdivided into one or more (preferably a plurality) of time slots ts.sub.m (m=1, . . . , M, where M is an integer number, and M>0). Each time slot ts.sub.m ranges from an initial instant t.sub.0(m) to a next instant t.sub.0(m+1) (excluded) which is the initial instant of the next time slot ts.sub.m+1, or:
ts.sub.m=[*t*.sub.0(m),*t*.sub.0(m+1)).
(45) Anyway, in embodiments overlapping time slots can be chosen, as well as one or more time slots can be absent during the predetermined observation time period TP. Also, the time slots ts.sub.m into which the observation time period TP is subdivided may have different lengths (time duration) from one another. Additionally, the observation time period TP may include just one time slot ts.sub.m of length equal to the length of the time period TP.
(46) For each one of the time slots ts.sub.m a respective Array of Presences **400***m* is computed that accounts for the presence of individuals in each zone z.sub.q of the RoI **300** during the time slot ts.sub.m. Therefore, a sequence or a set **500** of M Arrays of Presences **400***m*, as shown in FIG. **5**, is obtained that provides information about the presence of individuals in each one of the different zones z.sub.q of the RoI **300** during the observation time period TP.
(47) In one embodiment, commonly used values to which the observation time period TP and the time slots ts.sub.m are set can correspond to 24 hours and 15 minutes, respectively. Naturally, the scope of the present disclosure is not limited by any specific values selected for the observation time period TP and the time slots ts.sub.m.
(48) The computation of the presence of individuals on the zones of the RoI **300** and the generation

of the set **500** of Arrays of Presences **400***m* will be now described. The ROI **300** on which the system **100** has to perform the calculation of presence of individuals is completely tessellated by a tessellation **600**, as shown in FIG. **6**. For the sake of simplicity, a tessellation **600** with square tiles **605** all of equal size has been chosen, but for the purposes of the present disclosure, the shape of the tiles **605** of the tessellation **600** is irrelevant; for example, instead of square tiles, triangular or polygonal tiles can be used. The size of the tiles **605** is also not relevant.

(49) The tessellation **600** of the RoI **300** is independent from the coverage of the RoI **300** by the network cells of the mobile communication network **105**: there is no relationship between the tiles **605** and the network cells **105***b*. The tessellation **600** is for example inputted by the administrator of the system **100** through the Administrator Interface **120**.

(50) For the purposes of the present disclosure, the system **100** requests to the UE (possibly, to a subset thereof) that happens to be located in the RoI **300** to provide a periodical estimation of their position. The UE may for example exploit the GPS localization resources integrated therein for calculating their position estimates. The position estimates (hereafter also referred to as “position measurements”, since the position estimates are obtained by measuring one or more characteristics of radio signals received by the UE from the radio base stations, from GPS satellites or other GNSS, etc., depending on the localization technology adopted) are arranged as t-uples, e.g. triplets **700** of the type (UE_id, position (x,y), timestamp), depicted in FIG. **7**, where: the field UE_id **705** of the triplet **700** contains an indication of a univocal identifier of the UE that calculated and sent the position estimate. For the purposes of the present disclosure the value to be put in the field UE_id **705** may be selected between one or more among the International Mobile Equipment Identity—IMEI—, the International Mobile Subscriber Identity—IMSI— and the Mobile Subscriber ISDN Number—MSISDN—code; the field position (x,y) **710** of the triplet **700** contains an indication of the geographic position (longitude and latitude) estimated by the UE, i.e., the value of the position estimate, expressed in any system of geographic coordinates (e.g., WGS84), and the field timestamp **715** of the triplet **700** contains the indication of the time instant at which the position estimate was calculated/determined by the UE.

(51) The field UE_id **705** may also be absent. If present, this field allows optimizing some performances of the system **100**, as described later on. In one embodiment of the present disclosure, the value in the field UE_id **705** may be provided not in clear, as encrypted information in order to ensure the privacy of the UE owner. In other words, if the need arises, the information (i.e., the identity of an individual corresponding to the UE whose univocal identifier is specified in the field UE_id **705**) may be encrypted/decrypted by implementing a suitable encryption/decryption algorithm, or a cryptographic hash function such as for example the algorithm SHA256 described in “Secure Hash Standard (SHS)”, National Institute of Standards and Technology FIPS—180-4, Mar. 6, 2012. In this way, the identity of the individual is not in clear, being replaced by a non-intelligible pseudonym.

(52) The procedure implemented by the system **100** can be regarded as composed by two main phases, generically, but not necessarily, separated in time from each other.

(53) In a first phase of the procedure, cells-zones correspondence maps are constructed. This first phase is based on a set of position estimates (measurements) performed by a set of UE within the RoI **300**, exploiting any suitable geolocation technique available at the UE (e.g., the GPS). The time period MP during which such position estimates are carried out is subdivided into time slots, for example the same time slots ts.sub.m used to subdivide the time period TP during which the set **500** of Arrays of Presences **400***m* is to be calculated on the RoI **300**. For each time slot ts.sub.m of the time period MP, a cells-zones correspondence map is calculated: thus, a number M of cells-zones correspondence maps are calculated.

(54) In a second phase of the procedure, for each time slot ts.sub.m of the time period TP the presences on the zones z.sub.q into which the RoI **300** is subdivided are calculated, exploiting the cells-zones correspondence maps calculated in the same time slot ts.sub.m of the time period MP.

The set **500** of Arrays of Presences **400***m* is thus generated.
(55) The term “map” used herein should be not be intended as referring to a common geographic map, rather the term shall be intended as a tool that associates one or more numerical values (obtained by processing data provided by the mobile communication network) to entities representing portions of the territory (like the tiles **605** of the tessellation **600** or the zones zy of the subdivision into zones **300**).
(56) Referring to FIG. **8**, a procedure **800** according to an embodiment of the present disclosure for constructing cells-zones correspondence maps comprises four main steps: step **805**: configuration of those UE that will have to provide to the system **100** estimates of their position; step **810**: construction of general distribution maps (one general distribution map for each time slot ts.sub.m); step **815**: construction of cell-specific distribution maps (one map for each cell, for each time slot ts.sub.m), and step **820**: construction of cells-zones correspondence maps (one map for each time slot ts.sub.m).
(57) In greater detail, referring to FIG. **9**, in step **805** of the procedure **800** of configuration of the UE (the step **805** is denoted as **900** in FIG. **9**), the system **100** selects the UE to be included in a set of UE which will have to calculate and provide to the system **100** position estimates (step **905**). Then (step **910**), the Localization Manager **200**, in particular the Communication Manager **205**, sends to the selected UE configuration parameters to be used by the selected UE to perform and provide the position estimates; such configuration parameters may include: an indication of the time interval MP=[MP.sub.start, MP.sub.stop] during which the measurement campaign (to calculate the position estimates) will have to be carried out (e.g.: 24 hours from 00:00 to 24:00 of a specific day, etc.), an indication of the periodicity D with which the UE will have to estimate their position (e.g., every 30 seconds) and an indication (flag I/NI) of whether the UE will have to send the position estimates to the Position Manager **210** immediately after the position estimates have been calculated (flag I/NI set to value I) or the calculated position estimates can be sent by the UE at a later time (flag I/NI set to value NI), preferably before the end of the time interval MP of the measurement campaign. If the system **100** has to calculate a set **500** of Arrays of Presences **400***m* related to a time period TP included in the time period MP and such computation has to be carried out in real time, the system **100** configures the UE to immediately transmit their position estimates, as soon as they are calculated (flag N/I set to I). Differently, in case the system **100** does not need to compute the set **500** of Arrays of Presences **400***m* in real time, or in case the time period TP is not joined to the time period MP, the system **100** may configure the UE for a non-immediate transmission of their position estimates (flag N/I set to NI); this modality allows the UE to transmit two or more triplets **700** altogether, in a same block, or to exploit the establishment of communication links for other services, in both cases making the transmission more efficient, or a UE happening to be in an uncovered area can wait to return to a covered area and then transmit the position estimates.
(58) The selection of the UE to be inserted in the set of UE which will have to calculate and provide to the system **100** their position estimates and to which the configuration parameters have to be sent, can be accomplished in different ways. One possibility is to send the configuration parameters to all the UE which happen to be within the cells **105***b* of the mobile communication network **105** inside the RoI **300**, for example exploiting a broadcast message. Another possibility is that the administrator of the system **100**, possibly in agreement with the operator of the mobile communication network **105**, sends the configuration parameters only to a subset of those UE on the basis of one or more criteria, such as for example: UE of users which have given to the network operator their consent to activate their localization; UE of users which have given to the network operator or to the administrator of the system **100** their consent to take part to the measurement campaigns to be performed for the specific purposes of the system **100**; UE of users which, at the subscription of the service contract with the network operator, have declared a residence in a given area (e.g., inside the RoI **300** or inside a specific zone of the RoI **300**): UE of users in respect of

which the network operator has determined a residence or a working location in a certain area (based on any of the known techniques, e.g. the technique described in U.S. Pat. No. 9,706,363, disclosing a method for identifying and locating at least one relevant location visited by at least one individual within a geographical area served by a wireless telecommunication network the method including: identifying and clustering events that occurred within a predefined distance from each other and having a similar probability, computing a weight value for each cluster of events identified, comparing the weight value with a threshold weight value, if the weight value is equal to, or greater than, the threshold weight value, identifying the relevant location as belonging to a selected typology of relevant location, and providing an indication of the position of the at least one relevant location based on recorded position data of the events of the cluster, or if the weight value is lower than the threshold weight value, identifying the relevant location as not belonging to the selected typology of relevant location).
(59) The selection of the UE which will have to calculate and provide to the system **100** their position estimates is preferably made with the aim of complying with law prescriptions (e.g., concerning privacy of users) and/or to avoid unnecessary processing by the UE or the system **100** (for example, avoiding to include in the set of UE those UE which are expected to just quickly transit through the RoI **300**, and which, if selected to take part to the measurement campaign, would send position estimates also when they have moved (far) away from the RoI **300**). Additionally, the administrator of the system **100**, based on the available hardware/software resources (CPU, RAM, etc.) of the system **100**, can decide (e.g., to avoid overloading the system **100**, particularly the Localization Manager **200** and the Computation Engine **110**) to select, among all the possible UE, a suitably small number of UE. The UE can be selected in a random manner or based on specific requirements for the UE, for example based on the localization features and capabilities of the UE, e.g., in terms of achievable accuracy for the position estimate.
(60) Referring to FIG. **10**, the step **810** (denoted as **1000** in FIG. **10**) of construction of the general distribution maps (one general distribution map for each time slot ts.sub.m) may comprise the following operations.
(61) For each general distribution map to be constructed and for each tile **605** of the tessellation **600** in respect of each general distribution map, a counter n_pos is initialized to an initial value, e.g. to 0 (zero); the counter n_pos will be used to count the number of position estimates received by the selected UE falling in said tile in the considered time slot ts.sub.m (step **1005**).
(62) Then, starting from the time instant MP.sub.start and till the time instant MP.sub.stop of the time period MP during which the measurement campaign is carried out, the UE selected at step **905** estimate their own geographic position every I) seconds and send the calculated position estimates (in accordance to the timing specified by the setting of the flag I/NI) to the Position Manager **210**, e.g. in the form of triplets **700** as depicted in FIG. **7**; the Position Manager **210** stores the received triplets **700** in the Position Repository **215** (step **1010**).
(63) For each time slot ts.sub.m, at the end of the same, or, alternatively, at the end of the time period MP, the Position Manager **210** extracts from the Position Repository **215** all the received triplets **700** having a value in the respective field timestamp **715** included in the time slot ts.sub.m under consideration and sends the extracted triplets to the Computation Engine **110** (step **1015**). In case the time period TP is included in the time period MP and the system **100** has to process the data and create the set **500** of Arrays of Presences **400***m* in real time, the operation **1015** should start as soon as the timeslot ts.sub.m ends; otherwise the operation can be started after the end of the time period MP.
(64) The Computation Engine **110** receives from the Position Manager **210** the extracted triplets **700** and, for each received triplet, based on the geographic coordinates included in the field position (x,y) **710** thereof, it determines in which tile **605** of the tessellation **600** the considered triplet falls (step **1020**). The Computation Engine **110** increases of a predetermined amount (e.g., **1**) the counter n_pos corresponding to the determined tile **605** (step **1025**). In the case that the

geographic coordinates present in the field position (x,y) **710** of a triplet **700** under consideration are internal to the determined tile **605** there is a univocal assignment of the position of the corresponding UE to that tile (and the counter n_pos of that tile is increased); when instead the coordinates present in the field position (x,y) **710** of the triplet **700** are located on the boundary between two or more tiles **605** of the tessellation **600**, there are two possibilities: 1) a first possibility is to randomly select one of the two or more concerned tiles and to increase of the predetermined amount (e.g., 1) the counter n_pos corresponding to the randomly selected tile **605**; 2) another possibility is to increase the counters n_pos associated to all the two or more concerned tiles **605** of the predetermined amount (e.g., 1) divided by the number of concerned tiles **605**. The choice between one or the other of these two possibilities can for example be made by the administrator of the system **100**, through the Administrator Interface **120**.
(65) Once all the triplets **700** relating to the time slot ts.sub.m have been analysed by the Computation Engine **110**, the general distribution map is obtained that, for the considered time slot ts.sub.m, associates, with each tile **605** of the tessellation **600**, a value of the respective counter n_pos equal to the number of position estimates which have been received from the (selected) UE and which have geographic coordinates in the field position (x,y) **710** falling in said tile **605**.
(66) Then, the step **815** of construction of the cell-specific distribution maps for each time slot ts.sub.m starts. This step is schematically depicted in the flowchart of FIG. **11** where it is denoted **1100**.
(67) For each network cell c.sub.l of the mobile communication network **105** the Computation Engine **110** extracts from the Repository **115** the corresponding coverage area (step **1105**) and determines all the K tiles **605** of the tessellation **600** that belong to the coverage area of the cell c.sub.l (step **1110**). One way to determine that a tile **605** belongs to the coverage area of the cell c.sub.l is by assessing that the coordinates of the center of gravity of the tile **605** fall inside the coverage area of the cell c.sub.l, or that a given percentage (fraction) of the tile surface, for example a predetermined % value, falls inside the coverage area of the cell c.sub.l. The choice of the specific criterion/criteria for considering that a tile **605** belongs to the coverage area of the cell c.sub.l is for example made by the administrator of the system **100** through the Administrator Interface **120**.
(68) For each one of the K tiles determined at step **1110**, the Computation Engine **110** calculates a weight value weight % as the ratio between the numerical value of the counter n_pos for that tile **605** and the sum of the numerical values of the counters n_pos for all the K tiles:
(69) $\text{weight\%}_{l,i} = n\_pos_{l,i} \,/\, \text{.Math.}_{j=1}^{K}\, n\_pos_j$ (Equation1)
with the index i=1, . . . , K (step **1115**). The result of this operation is, for each time slot ts.sub.m, the cell-specific distribution map for the cell c.sub.l, which establishes a correspondence between each tile **605** of the tessellation **600** and a weight value calculated as in Equation 1. Such a weight value can be regarded as the probability that a user of a UE that is located in the coverage area of the cell c.sub.l is, in the generic time slot ts.sub.m, specifically located in the i-th tile **605** of the tessellation **600**. By repeating these operations for all the cells of the mobile communication network **105** a set of cell-specific distribution maps is obtained for every time slot ts.sub.m, one for every cell.
(70) When the time period MP lasts 24 hours or, more generally, a few days (the duration of the time period MP may depend on how often the mobile communication network operator modifies the coverage of the network) the system **100** can be configured by the system administrator in such a way that the operations **1105** and **1110** are performed only once, at the time the triplets **700** of the first time slot ts of the time period MP are processed. The tiles **605** identified at that time are used, without changes, also during the processing performed for the other time slots from ts.sub.2 to IS.sub.M.
(71) At this point, the step **820** of constructions of cells-zones correspondence maps (one for each

time slot ts.sub.m) commences, as schematized by the procedure **1200** in the flowchart of FIG. **12**.
(72) For each zone z.sub.q of the subdivision in zones of the RoI **300** the K′ tiles **605** of the tessellation **600** that belong to the considered zone z.sub.q are identified (step **1205**). One way to determine that a tile **605** belongs to the zone z.sub.q is by assessing that the coordinates of the center of gravity of the tile **605** fall inside the zone z.sub.q, or that a given percentage (fraction) of the tile surface, for example a predetermined % value, falls inside the zone z.sub.q. The choice of the specific criterion/criteria for considering that a tile **605** belongs to the zone z.sub.q is for example made by the administrator of the system **100** through the Administrator Interface **120**. For simplifying the configuration and the management of the system **100**, the specific criteria adopted (coordinates of the center of gravity of the tile **605** inside the zone z.sub.q, a given percentage of the tile **605** inside the zone z.sub.q or any other possible criterion) for identifying such K′ tiles may be the same adopted in step **1110** for identifying the K tiles belonging to the coverage area of each cell of the mobile communication network **105**.
(73) In addition, step **1205**, similarly to step **1110**, can be performed just once, at the time of processing of the position estimates of the first time slot ts.sub.1 of the time period MP.
(74) Then, for each time slot ts.sub.m and for each cell c.sub.l of the mobile communication network **105**, by exploiting the cell-specific distribution map for that cell, calculated in the preceding step **815** and referring to the time slot ts.sub.m (steps **1105-1115**), the weight (P.sub.t.fwdarw.q).sub.m of the cell c.sub.l on the zone z.sub.q is calculated by summing up the weights of the K′ tiles identified at step **1205** and obtained from the map (step **1210**), i.e.:
(75) $(P_{l\,.\text{fwdarw.}\,q})_m = .\text{Math.}_{i=1}^{K'}\ (\text{weight}\%_{l,i})_m\ .$ (Equation2)
(76) Such weight (P.sub.t.fwdarw.q).sub.m can be regarded as the probability that a user of a UE that is located in the coverage area of the cell c.sub.l is, in the time slot ts.sub.m, located in the specific zone z.sub.q among all the zones of the partition in zones of the RoI **300**.
(77) By repeating this operation in every time slot ts.sub.m and for all the cells c.sub.l of the mobile communication network **105**, the cells-zones correspondence map is obtained, which is the map in which every zone z.sub.q of the subdivision in zones of the RoI **300** is associated with the total weight of each cell c.sub.l of the mobile communication network **105** on such a zone in the time slot ts.sub.m.
(78) The cells-zones correspondence maps, relating to the time slot ts.sub.m of the time period MP, are exploited for calculating, in the corresponding time slot ts.sub.m of the time period TP, the presences on the zones z.sub.q in which the RoI **300** is subdivided, and thus the set **500** of Arrays of Presences **400***m*. The operations are schematized as **1300** in the flowchart of FIG. **13** and are described herebelow.
(79) At the end of every time slot ts.sub.m, all the event records er.sub.v having a timestamp falling in that time slot are extracted from the Repository **115** (step **1305**).
(80) For each event record er.sub.v, the cell c.sub.l in which the event occurred is identified (step **1310**) and, exploiting the cells-zones correspondence map for that cell, the K″ zones z.sub.q in which that cell has a weight (P.sub.t.fwdarw.q).sub.m different from zero are determined (more generally, the cell has a weight greater than a predetermined cell weight value and this predetermined cell weight value is is preferably set by the administrator of the system **100** through the Administrator Interface **120**) (step **1315**). The value in the field of the Array of Presences **400***m* corresponding to the determined zones is increased of an amount equal to (P.sub.t.fwdarw.q).sub.m. By repeating steps **1310** to **1320** for all the event records er.sub.v extracted from the Repository **115** at step **1305**, the final Array of Presences **400***m* is obtained, i.e. the array containing the number of presences of individuals in each zone of the RoI **300** in the time slot ts.sub.m.
(81) The Array of Presences **400***m* is provided in output by the system **100**, to the user of the system **100**, through the User Interface **125**.
(82) By repeating the operations in the flowchart **1300** for all the M time slots of the time period

TP, the complete set **500** of Arrays of Presences **400***m* is obtained.
(83) After having provided the set **500** of Arrays of Presence **400***m*, the operation of the system **100** ends.
(84) In other embodiments, the method may include different steps, or some steps may be performed in a different order or in parallel.
(85) In case, during operation **1305**, it comes out that in the time slot ts.sub.m there are more than one event records er.sub.v related to a same UE, before performing operation **1310** it may be preferable to select just one event record er.sub.v, to avoid that a same UE, and thus a same user, is counted twice or more within a same time slot ts.sub.m; for example, it is possible to select the first event record er.sub.v, or the last event record er.sub.v related to that UE, or an event record er.sub.v among the two or more event records related to that UE is selected randomly among the event records related to that UE. The selection criterion may be set by the administrator of the system **100**, through the Administration Interface **120**. Instead, in the case that during the time slot ts.sub.m a UE does not have any related event record er.sub.v, even if in at least one preceding time slot that UE had at least one event record er.sub.v, the Computation Engine **110**, before performing operation **1310**, may decide to exploit, for that UE, the last, more recent event record er.sub.v available for the preceding time slots (in order to take into account, in the count of presence of individuals, also users that in a certain time slot did not perform any activity but can be reasonably assumed to still be in the previous position). Also this operation is decided by the administrator of the system **100** through the Administrator Interface **120**.
(86) If the Array of Presences **400***m*, upon request of the user of the system **100** through the User Interface **125**, has to be generated in real-time, the operations globally denoted as **1300** have to be carried out as soon as the time slot ts.sub.m ends, so as to ensure that all the event records er.sub.v with timestamp within that time slot are taken into consideration. When instead the system **100** is not requested to generate the Array of Presences **400***m* in real-time, the operations **1300** can be carried out at any time, for example even at the end of the time period TP. The timing for carrying out the operations can be decided by the administrator of the system **100** through the Administrator Interface **120**.
(87) If the system **100** is exploited for calculating the presence of individuals in two or more RoIs **300** at a time, during operation at step **1015** the Position Manager **210** can look and take into consideration, in addition to the field timestamp **715** of the triplet **700** containing the position estimate, also the identifier, contained in the field UE_id **705** of the triplet, of the UE that calculated the position estimate (if the field UE_id **705** is present) and retrieve from the Position Repository **215** only the position estimates coming from those UE that, with the operation at step **905**, have been identified to be in a certain one of the RoIs **300**.
(88) If, while performing the operations at step **1015** for the time slot ts.sub.m under analysis, it is determined that no useful triplets **700** are stored in the Position Repository **215**, or the available triplets **700** are in a number less than a minimum threshold number, the Position Manager **210** may extract from the Position Repository **215** also the triplets **700** included in the immediately adjacent (preceding and successive) time slots, until the overall number of extracted position estimates reaches the threshold number. All these triplets are then sent to the Computation Engine **110** for being processed, as if such triplets **700** were all internal to the time slot ts.sub.m under consideration. This option and the minimum threshold number can be decided and configured by the administrator of the system **100** through the Administration Interface **120**.
(89) In order to reduce the processing time of the system **100** or to reduce the need of communication bandwidth for the communication between the Position Manager **210** and the Computation Engine **110** when the two modules are physically separated from each other, the administrator of the system **100** can decide to reduce the number of, e.g. to decimate, the position estimates **700** to be processed by the Position Manager **210** before the transmission to the Computation Engine **110** (operation at step **1015**), or to be processed by the Computation Engine

110 before starting the processing (operations at step **1020**). In this way, just one triplet **700** (i.e., just one position estimate) is taken out of a number n of triplets **700** (i.e., n position estimates). Such a reduction in number of the position estimates to be subject to processing can be operated on a per-user (per-UE) basis, in case the field UE_id **705** is present in the triplets **700** (this means that one measurement is retained for each value of the field UE_id), or globally on all the position estimates (this means that one measurement is retained every n measurements, irrespective of the UE-identified by the value in the field UE_id—that performed the measurements).

(90) For similar reasons, the Position Manager **210** or the Computation Engine **110** can be configured by the administrator of the system **100** (through the Administrator Interface **120**) to calculate, from the set of position estimates extracted during the operations at step **1015**, an average position for each UE, if the field UE_id **705** is present, and to exploit this single average position in the operations at step **1020**. Also this option can be decided by the administrator of the system **100**.

(91) The set of measurements (position estimates) performed during a measurement campaign MP can be exploited without changes for calculating the set **500** of Arrays of Presences **400***m* related to more than one observation time periods TP. In other words, when a user of the system **100** requests to the system **100** the calculation of the set **500** of Arrays of Presences **400***m* on a certain time period TP, before starting a measurement campaign on a certain time period MP, the user may ascertain if in the system **100**, particularly in the Position Repository **215**, there are already stored triplets **700** obtained in a preceding time period MP on the RoI **300** of interest for the user, and decide to exploit the already available triplets **700** on the subdivision in zones of the RoI of that user. In this case, the system **100** starts directly with the operations at step **1015**. Even more, if the tessellation and/or the subdivision in zones of the RoI of interest for that user of the system **100** are the same to that/those used previously, the user of the system **100** may decide to reuse the general distribution maps and the cells-zones correspondence maps already calculated previously. The processing burden for calculating the new set **500** of Arrays of Presences **400***m* is consequently reduced.

(92) The solution according to the present disclosure is independent from the specific localization technique employed by the UE for estimating their positions: it is for example possible to employ a technique based on mobile radio technology, standard or not (Cell-Id, TOA, based on radio signal strength measurements, etc.), or a technique based on short-range communication capabilities of the UE (e.g., Wi-Fi, Bluetooth, etc.) or a satellite-localization technique (GPS, Galileo, etc.).

(93) In an embodiment of the present disclosure, the UE can send, within the triplets **700**, particularly within the field position (x,y) **710** of the triplets **700**, also an indication of the uncertainty, or of the accuracy, associated with the estimated position. The Computation Engine **110**, when performing the operations at step **1020**, can decide to use only those position estimates that are accompanied by a value of uncertainty lower than a predefined uncertainty threshold, for example set by the administrator of the system **100** through the Administrator Interface **120**. The uncertainty threshold can depend on the characteristic size of the generic tile **605** of the tessellation **600** (for example, in case of square tiles **605**, the uncertainty threshold can be set equal to half the length of the tile side; for circular tiles, the uncertainty threshold can be set equal to the radius of the circumference delimiting the tile).

(94) In an embodiment of the present disclosure, the operations at step **810** can be cooperatively carried out also with the participation of those UE that are not capable of estimating their position (either because they do not have installed a suitable software or because the software available to them always provided results affected by an a-priori unacceptable uncertainty). These can be UE that are only capable of measuring the characteristics of the radio signal received from the radio base stations **105***a* (e.g., serving radio base station, received radio signal power, adjacent radio base stations, etc.). The Position Manager **210**, upon receiving these measurements, can calculate the position of the UE exploiting a localization algorithm and store the results in the Position Repository **215** (step **1010**).

(95) The system **100** can be implemented by means of an app/server architecture, which provides for having an app installed on the UE which dialogues with a remote server using, at high level, Internet protocols and, at a lower level, the User-plane of the mobile communication network **105**. Alternatively, the solution according to the present disclosure can be actuated exploiting, at least in part, messages and functions of the MDT (Minimization of Drive Tests) technology, introduced in the Release 10 of the 3GPP standard for UMTS and LTE (see e.g. the 3GPP TS 37.320). For example, in case of immediate transmission (flag I/NI set to I) of the position estimates, it is possible to exploit the functionality called “Immediate MDT”, whereas in case of delayed transmission (flag I/NI set to NI) it is possible to exploit the functionality called “Logged MDT”.

Claims

1. A method, implemented by a data processing system, for estimating presence of individuals on a territory based on data derived from a mobile communication network comprising a plurality of network cells covering said territory, wherein said individuals are users of mobile communication terminals interact with the mobile communication network and located in said territory, the method comprising: subdividing a geographic area of interest in geographic zones; subdividing the geographic area of interest in geographic area portions; causing mobile communication terminals served by the mobile communication network and situated in the geographic area of interest to calculate respective geographic position estimates and provide the calculated geographic position estimates to the data processing system; distributing an overall number of geographic position estimates, received from the mobile communication terminals, on the geographic area portions by assigning to each geographic area portion a respective number of geographic position estimates corresponding to geographic positions estimates falling within said geographic area portion; for each of said at least one network cell: determining, among said geographic area portions, covered geographic area portions falling within a coverage area of said network cell; assigning to each one of the determined covered geographic area portions a respective weight which depends on said respective number of geographic position estimates compared to an overall number of geographic position estimates falling within all said determined covered geographic area portions; and generating a correspondence map establishing a correspondence between said network cells and said geographic zones, said generating the correspondence map comprising, for each of said network cells, calculating a cell weight of the network cell on each geographic zone by determining the geographic area portions belonging to each of said zones and summing the respective weights of the geographic area portions identified as belonging to said each of said zones; receiving from the mobile communication network and storing in a repository indications of events of interaction between the mobile communication terminals and the network cells of the mobile communication network; identifying, for each one of the stored indications of events of interaction, the network cell in which the event of interaction occurred; using the correspondence map, building an array of presences in which each of said geographic zones has assigned thereto a respective indication of a respective number of events of interaction occurred in the identified network cells which have, in said each of said geographic zones, a cell weight greater than a predetermined cell weight value; and providing said array of presences in output to said data processing system for the use, by a user of the data processing system, for estimating presence of individuals on a territory.
2. The method of claim 1, wherein said events of interaction comprise at least one among: interactions at the power-on and/or power-off of the mobile communication terminals; interactions at location area update procedures; interactions at the re-entrance of the mobile communication terminals into a covered area, covered by the mobile communication network, from an area not covered by the network; interactions at the switch of the mobile communication terminals from a mobile network technology to another; interactions at incoming/outgoing calls; interactions at sending/receiving SMS and/or MMS; and interactions at Internet access.

3. The method of claim 1, wherein said building an array of presences comprises building a set of arrays of presences, each array of presences of said set being related to a respective time slot of a plurality of time slots in which an observation time period is subdivided.
4. The method of claim 1, wherein said causing mobile communication terminals served by the mobile communication network and situated in the geographic area of interest to calculate respective geographic position estimates and provide the calculated geographic position estimates to the data processing system comprises selecting, among all the mobile communication terminals served by the mobile communication network and situated in the geographic area of interest, a subset of selected mobile communication terminals according to a selection criterion comprising at least one among: mobile communication terminals of users giving consent for activating their geographic localization; mobile communication terminals of users giving consent for measurement campaigns calculating the presence of individuals on a geographic territory; mobile communication terminals of users which, at subscription, have declared a residence in a certain area inside the geographic area of interest; and mobile communication terminals of users in respect of which the operator of the mobile communication network has determined a residence or a working location in a certain area inside the geographic area of interest.
5. The method of claim 4, comprising configuring the subset of selected mobile communication terminals, said configuring comprising providing to the selected mobile communication terminals at least one among: indicating a time interval during which the selected mobile communication terminals calculate the respective geographic position estimates; indicating a periodicity with which the selected mobile communication terminals calculate the respective geographic position estimates; and indicating whether the selected mobile communication terminals providing to the data processing system the respective calculated geographic position estimates immediately after calculating the respective geographic position estimates or providing to the data processing system after calculating the respective geographic position estimates.
6. The method of claim 1, wherein determining covered geographic area portions falling within a coverage area of that network cell comprises: obtaining a coverage area of that network cell; and determining which of said geographic area portions belong to the coverage area of the network cell, wherein said determining comprises at least one among: assessing which of said geographic area portions have geographic coordinates of a center of gravity thereof falling inside the coverage area of the network cell; and assessing which of said geographic area portions fall, for a given percentage of their area, inside the coverage area of the network cell.
7. The method of claim 1, wherein said determining the geographic area portions belonging to each of said zones comprises one among: assessing which of said geographic area portions have geographic coordinates of a center of gravity thereof falling inside the zone; and assessing which of said geographic area portions fall, for a given percentage of their area, inside the zone.
8. The method of claim 1, wherein said assigning to each geographic area portion a respective number of geographic position estimates corresponding to geographic positions estimates falling within said geographic area portion comprises, in case a geographic position estimate falls on a boundary between at least two geographic area portions either: randomly selecting and assigning one of the at least two geographic area portions to that geographic position estimate, or assigning to each of said at least two geographic area portions a fraction of said geographic position estimate equal to 1 divided by the number of said at least two geographic area portions.
9. The method of claim 1, wherein said calculated geographic position estimates are provided in the form of t-uples, wherein each t-uple comprises: a field containing an indication of the geographic position estimated by the mobile communication terminal, expressed in any system of geographic coordinates, a field containing an indication of the time instant at which the geographic position estimate has been calculated, and a field containing an indication of a univocal identifier of the mobile communication terminal that calculated and sent the geographic position estimate.

10. A data processing system comprising modules configured to perform, when operated, the method of claim 1.